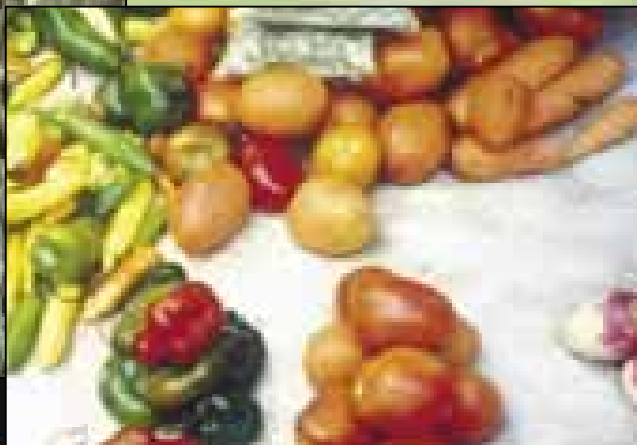


Fertilizer use by crop in Pakistan



Fertilizer use by crop in Pakistan

**Land and Plant Nutrition Management Service
Land and Water Development Division**

**FOOD AND AGRICULTURE ORGANIZATION OF THE UNITED NATIONS
Rome, 2004**

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Food and Agriculture Organization of the United Nations
Viale delle Terme di Caracalla
00100 Rome, Italy
Tel.: +(39) 06 57051
Fax: +(39) 06 57053360
E-mail: land-and-water@fao.org
Web site: www.fao.org

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Abstract

The total population of Pakistan is about 150 million, of which 67 percent live in rural areas and depend mainly on agriculture. It is growing at an annual rate of almost two percent. The total cropped area is 22 million ha, of which 18 million ha (80 percent) are irrigated. According to the 1985/90 soil survey there are 1.8 million ha of saline soils but the survey does not indicate what proportion is due to poor irrigation practices. Waterlogging is an important issue.

There are over five million farms in the country, 81 percent of which have less than five ha. Only seven percent of the farms have a size of over 20 ha but they account for forty percent of the farmed area.

Agriculture accounts for 24 percent of the growth domestic product (GDP) of Pakistan, employs 48 percent of the labour force and contributes about 60 percent to export earnings. GDP growth continues to depend on crop performance. About 32 percent of the population lives below the poverty level.

The fertilizer sector has been privatized and deregulated. Farm price subsidies have been eliminated and prices decontrolled. Almost ninety percent of fertilizers are distributed by the private sector through a network of some 8 000 dealers.

A large amount of data is available on crop responses to fertilizer and the use of fertilizers has been found to be profitable on all crops. Most of the fertilizer is used on irrigated wheat, cotton, sugar cane and rice. On these crops the nitrogen application rate is close to 80 percent of the recommendations, compared with about 40 percent or less, depending on the crop, in the case of phosphate. Scarcely two percent of farmers apply potash; the quantities used are applied to fruit and vegetable crops and sugar cane. Micronutrient deficiencies are common but less than five percent of the farmers apply micronutrient fertilizers.

Total food crop production in Pakistan increased from 10 million tonnes in 1970/71 to about 25 million tonnes in 2002/03. Fertilizer consumption increased 13-fold during the same period, to reach three million tonnes of total nutrients. However, the use of nutrients is unbalanced. The current $N:P_2O_5:K_2O$ ratio is 1:0.28:0.01. Growth in food production and hence in fertilizer use will continue due to investments in irrigation projects and increased food demand. What

needs to be done is known but not implemented. Fertilization practice is far from the recommendations with consequent loss of yield, financial waste and environmental contamination.

Preface

This study, commissioned by the Food and Agriculture Organization of the United Nations (FAO), is one of a series of publications on fertilizer use on crops in different countries.

The aim of the series is to examine the agro-ecological conditions, the structure of farming, cropping patterns, the availability and use of mineral and organic plant nutrients, the economics of fertilizers, research and advisory requirements and other factors that have led to present fertilizer usage. The reports examine, country by country, the factors that will or should determine the future development of plant nutrition.

During the past two decades, increasing attention has been paid to the adverse environmental impact of both the under use and the over use of plant nutrients. The efficient use of plant nutrients, whether from mineral fertilizers or from other sources, involves the shared responsibility of many segments of society, including international organizations, governments, the fertilizer industry, agricultural research and advisory bodies, traders and farmers. The publications in the series are addressed to all these parties.

Fertilizer use is not an end in itself. Rather it is a means of achieving increased food and fibre production. Increased agricultural production and food availability can, in turn, be seen as an objective for the agricultural sector in the context of contributing to the broader macroeconomic objectives of society. A review of the options available to policy-makers is given in the FAO/International Fertilizer Industry Association (IFA) 1999 publication entitled *Fertilizer Strategies*.

The contents of the series studies differ considerably from country to country, in view of their different structures, histories and food situation. But in each case the aim is to arrive at a better understanding of the nutrition of crops in the country concerned.

Abbreviations and symbols

CPR	Crop Production Region
FAO	Food and Agriculture Organization of the United Nations
f.o.b.	Free on board
FYM	Farm yard manure
IFA	International Fertilizer Industry Association
IMPHOS	World Phosphate Institute
Karaz	Underground channel for irrigation
Kharif	Summer crop season: April to September
NFDC	National Fertilizer Development Centre
NFWP	North West Frontier Province
PARC	Pakistan Agricultural Research Council
Rabi	Winter crop season: October to March
SCARP	Salinity Control and Reclamation Project
VCR	Value Cost Ratio
Zn	Zinc
AS:	Ammonium sulphate
CAN:	Calcium ammonium nitrate
DAP:	Diammonium phosphate
MOP:	Muriate of potash (potassium chloride)
NP:	Compound fertilizer containing nitrogen and phosphate
NPK:	Compound fertilizer containing nitrogen, phosphate and potash
SOP:	Sulphate of potash
SSP:	Single superphosphate
N:	Nitrogen
P ₂ O ₅ or P:	Phosphate*
K ₂ O or K:	Potash*

* Phosphate and potash may be expressed as their elemental forms P and K or as their oxide forms P₂O₅ and K₂O. Nitrogen is expressed as N. In this study phosphate and potash are expressed in their oxide forms.

Chapter 1

Introduction

Pakistan has a total land area of 80 million ha. Of this, 22 million ha are used for crop production. About 18 million ha (80 percent) of the cultivated land is irrigated while the remainder is under dry farming. The range land, which covers over 50 percent of the total area of Pakistan, is a potential source of livestock development in the country. The total population is about 150 million. During the period from 1970/71 to 2002/03, the cultivated area increased from 16.62 million ha to 22.15 million ha, with an annual growth rate of 0.9 percent. During the same period, the population increased from 65 million to 150 million, registering a growth of 2.8 percent per annum. In consequence the per capita land availability decreased from 0.25 ha in 1970/71 to 0.15 ha in 2002/03. With a projected annual growth of 1.9 percent, the population will be around 190 million by the year 2015.

Agriculture is the mainstay of Pakistan's economy. It accounts for 24 percent of the GDP and employs 48.4 percent of the total labour force. Agriculture contributes to growth as a supplier of raw materials to industry as well as a market for industrial products and also contributes 60 percent to Pakistan's export earnings. About 67 percent of the country's population live in rural areas and are directly linked with agriculture for their livelihood. The agricultural sector grew at an average rate of 4.5 percent per annum during the decade of the 1990s. Agricultural production fell in 2000/01 and 2001/02 due to serious droughts, but grew by 4.2 percent in 2002/03. The crop sector accounts for about 60 percent of agriculture's contribution to the GDP, the livestock, fishery and forestry sectors for 40 percent.

There are two principal crop seasons in Pakistan namely "Kharif", with sowing beginning in April and harvest between October and December and "Rabi" beginning in October-December and ending in April-May. Rice, sugar cane, cotton, maize and millet are Kharif crops, while wheat, gram, tobacco, rapeseed, barley and mustard are Rabi crops.

The average per capita income was US\$492 in 2002/03. About 32 percent of the population live below the poverty level, based on a caloric norm of 2 350 calories per adult per day. This norm represented a per capita expenditure of

Rs.741 per month in 2000/01. According to this definition about 38.7 percent of the poor live in rural areas. This poverty level persists not because of inadequate food availability, but because poor people cannot afford to buy food.

The total food crop production in Pakistan increased from 10 million tonnes in 1970/71 to about 25 million tonnes in 2002/03. This was achieved due to technological developments in varieties, water availability, and fertilizer use.

Nitrogenous fertilizers were introduced in Pakistan in 1952, phosphorus seven years later in 1959/60, and potassium another seven years later in 1966/67. Fertilizer use gained momentum after 1966/67, when high yielding varieties of cereal crops were introduced.

Most fertilizers are used on irrigated crops. In rainfed areas use is still very low.

The objective of this paper is to review fertilizer use by crop in the crop production regions of Pakistan, fertilizer supply and the economic profitability of fertilizer use.

Chapter 2

Agro-ecological zones and crop production regions

Pakistan is situated between the latitudes of 24° and 37° north and longitudes of 61° to 75° east, stretching over 1 600 kilometres from north to south and 885 kilometres from east to west, with a total area of 796 095 square kilometres. It has a subtropical and semi-arid climate. The annual rainfall ranges from 125 mm in the extreme southern plains to 500 to 900 mm in the sub-mountainous and northern plains. About 70 percent of the total rainfall occurs as heavy downpours in summer from July to September, originating from the summer monsoons, and 30 percent in winter. Summers, except in the mountainous areas, are very hot with a maximum temperature of more than 40 °C, while the minimum temperature in winter is a few degrees above the freezing point.

AGRO-ECOLOGICAL ZONES

Basically, the country has been divided into ten agro-ecological zones based on physiography, climate, land use and water availability (Figure 1).

Zone I

Indus Delta. The climate is arid tropical marine. The mean monthly summer rainfall is 75 mm and winter rainfall less than 5 mm. The mean daily temperature is between 34 °C and 40 °C in summer and between 19 °C and 20 °C in winter. The soils are clayey and silty. Rice, sugar cane, banana and pulses are the major crops.

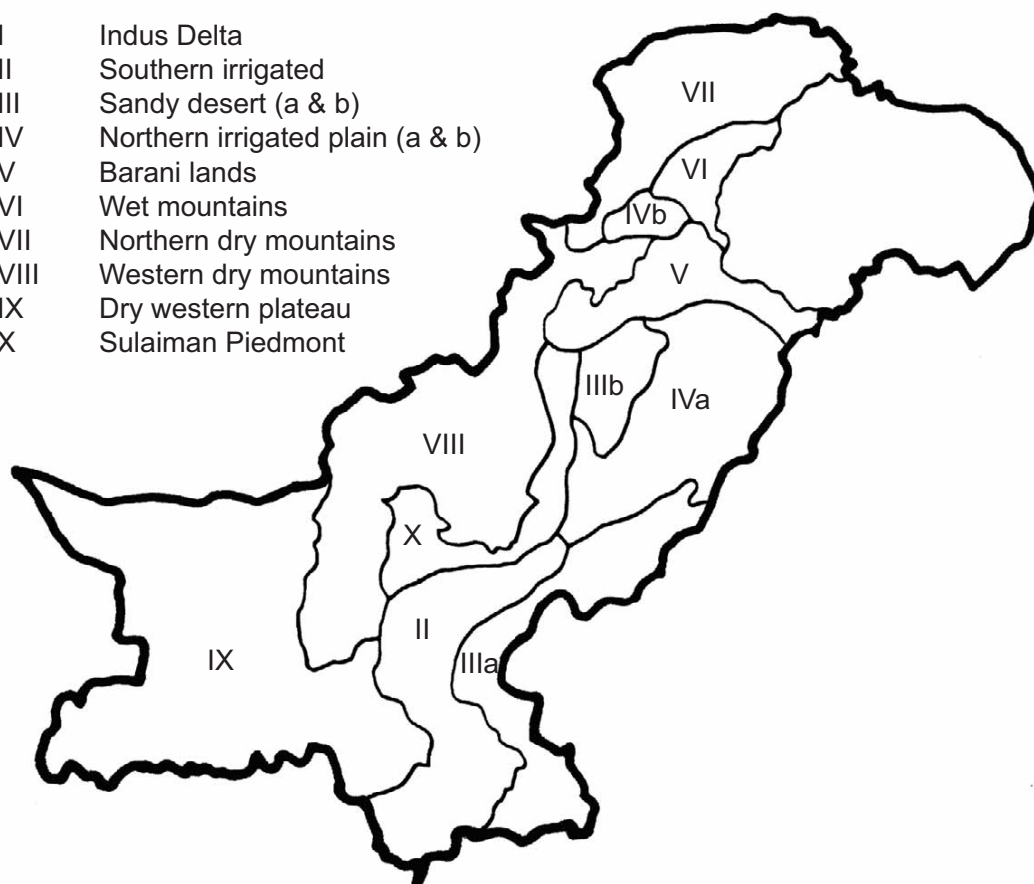
Zone II

Southern Irrigated Plain, the Lower Indus Plain. The climate is arid and sub-tropical. The mean monthly summer rainfall is 18 mm in the north and 45 to 55 mm in the south. The soils are silty and sandy loam but the upper areas of the flood plain are calcareous loamy and clayey. Cotton, wheat and sugar cane are grown on the left bank of the Indus and rice, wheat and gram on the right bank.

FIGURE 1

Agro-ecological zones

- I Indus Delta
- II Southern irrigated
- III Sandy desert (a & b)
- IV Northern irrigated plain (a & b)
- V Barani lands
- VI Wet mountains
- VII Northern dry mountains
- VIII Western dry mountains
- IX Dry western plateau
- X Sulaiman Piedmont



Source: PARC, 1980

Zone IIIa

Sandy Desert (a). The maximum rainfall is 300 mm. The soils are sandy and loamy fine sand. The land is used for grazing.

Zone IIIb

Sandy Desert (b) – sand ridges and dunes. The rainfall is between 300 and 350 mm. The soils are sandy and loamy fine sand. The land is used for grazing.

Zone IVa

Northern Irrigated Plain (a) – Flood Plains and Bar Uplands. The climate is semi-arid to arid. The mean annual rainfall is 300 to 500 mm in the east and 200 to 300 mm in the southwest. The soils are sandy, loam-clay and loam. The canal

irrigated crops are wheat, rice, sugar cane, oilseed and millets in the north and wheat, cotton, sugar cane, maize, citrus and mangoes in the centre and south.

Zone IVb

Northern Irrigated Plain (b) – alluvial valleys of Peshawar and Mardan. The climate is semi-arid. The mean monthly rainfall is 20 to 30 mm. The soils are silty clays and clay loams. The main crops are sugar cane, maize, tobacco, wheat, berseem, sugar beet and orchards.

Zone V

Barani (rainfed) Lands. This covers the Salt Range and the Potwar Plateau. In the North the mean monthly rainfall is 200 mm in summer and 35 to 50 mm in winter. The climate in the Southern part is semi-arid and hot. The mean monthly rainfall is 85 mm in summer and 30 to 45 mm in winter. The main crops are wheat, millet, oilseed and pulses.

Zone VI

Wet Mountains – High Mountains. The mean monthly rainfall is 235 mm in summer and 116 mm in winter. The soils consist of silt loams to silty clays. A small area is under rainfed agriculture but most of it is under forest.

Zone VII

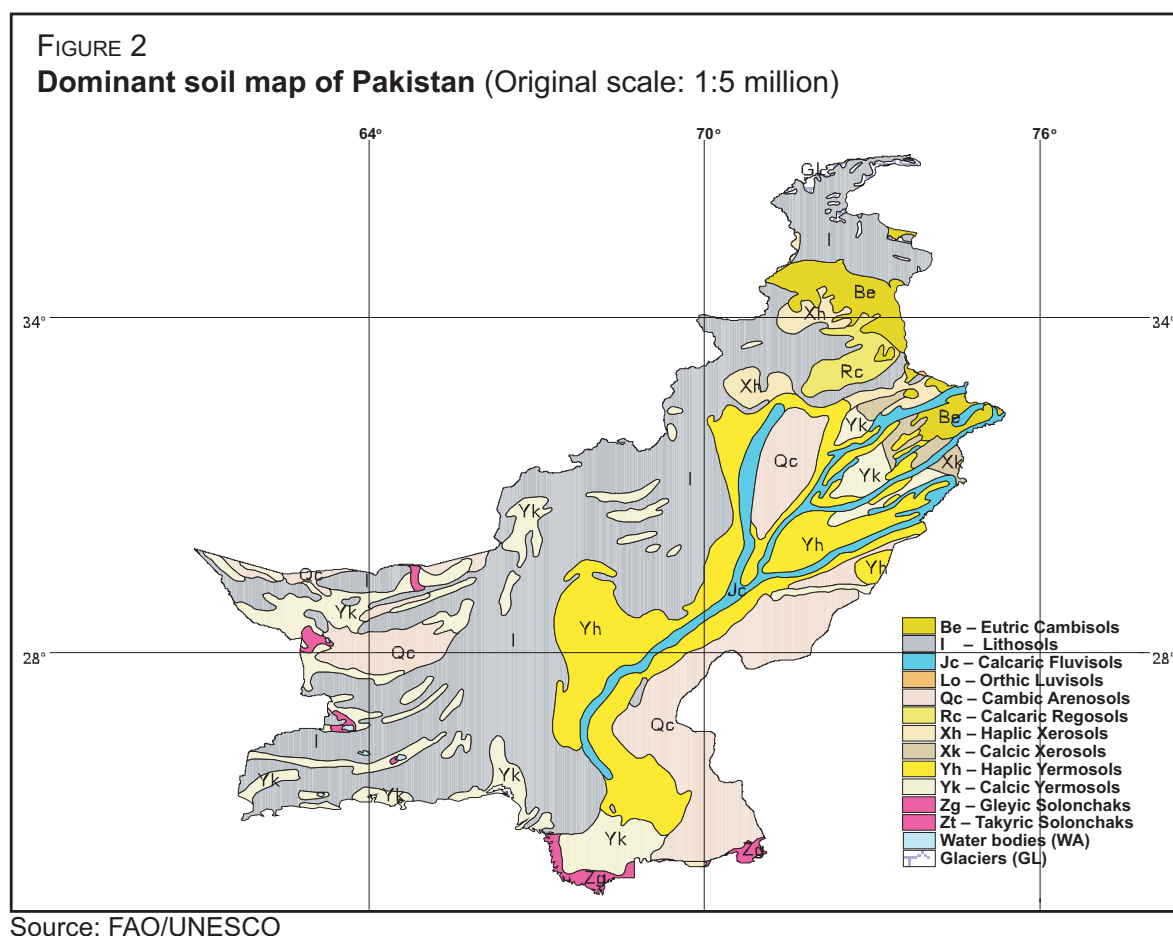
Northern Dry Mountains. The mean monthly rainfall is 25 to 75 mm in winter and 10 to 20 mm in summer. The valley soils are deep and clayey. Most of the area is used for grazing.

Zone VIII

Western Dry Mountains. They are composed of barren hills with steep slopes. The mean monthly rainfall is 95 mm in summer and 63 to 95 mm in winter. The soils in the valleys are deep and loamy. Most of the land is used for grazing. On part of the loamy soils wheat and fruit crops are grown.

Zone IX

Dry Western Plateau – mountainous areas. The mean monthly rainfall is 37 mm in summer. The coastal belt receives a sea breeze. The land is used mainly for



grazing. Melons, fruit crops, vegetables and wheat are grown where water is available.

Zone X

Sulaiman Piedmont – plains of the Sulaiman Range. The climate is arid and hot. The mean monthly rainfall is less than 15 mm. Irrigation relies on floods of the hill torrents. Wheat, millet and gram are the main crops.

Figure 2 shows the dominant soils of Pakistan.

IRRIGATION

The 4 000 year old Indus civilization has its roots in irrigated agriculture. The abandoned creeks of the meandering river, inundated during the flood season, are supposed to have served as the irrigation channels of the ancient systems. The first controlled all-year irrigation began in 1859 with the completion of the

Upper Bari Doab Canal emanating from the Madhopur headworks on the Ravi river. The Sukkur barrage, completed in 1932, is considered as the first modern hydraulic structure on the downstream Indus river.

Pakistan can be divided into three hydrological units:

- The Indus basin, covering more than 566 000 km² (or 71 percent of the territory), comprising the whole of the provinces of Punjab, Sindh and NWFP as well as the eastern part of Balochistan.
- The Karan desert in the west of Balochistan (in the west of the country), which is an endorheic basin covering 15 percent of the territory.
- The arid Makran coast along the Arabian Sea covering 14 percent of the territory in its southwestern part (Balochistan province).

Total water withdrawal in 2001 was estimated at 166 km³, of which 95.6 percent for agricultural purposes (4.4 percent is withdrawn for domestic use and another 1.6 percent for industrial use). Groundwater abstraction for agriculture has been roughly estimated at 63 km³/year through more than 500 000 tubewells. However, in some areas, development appears to have reached the point where groundwater is being mined. Most urban and rural water is supplied from groundwater. Over 50 percent of the village water supply is obtained through hand pumps installed by private households. In saline groundwater areas, irrigation canals are the main source of domestic water.

According to the usual Pakistani classification irrigation consists of:

- Government canals: 6.38 million ha in 2001/02, of which 58 percent in the Punjab and 29 percent in Sindh province;
- private canals: 0.43 million ha, of which 81 percent are in NWFP;
- tubewells: 3.45 million ha, of which 82 percent are in Punjab province;
- open wells: 0.2 million ha, of which 55 percent are in Punjab province;
- canals and tubewells: 7.24 million ha, all of them in Punjab province;
- other means: 0.18 million ha.

The total irrigated area is 18 million ha. About 4 million ha is rainfed. The main irrigated crops are wheat, rice, sugar cane and cotton. Owing to inadequate water availability in winter (storage capacity is too small) and at the beginning and end of summer, cropping intensity is exceptionally low. According to a World

Bank report, Pakistan does not have enough reservoir capacity in its irrigation system to store seasonal waters.

According to the Soil Survey of Pakistan (Mian and Javed, 1993), 2.8 million hectares of irrigated land is affected by salinity ranging from patchy salinity to dense saline sodic soils.

CROP PRODUCTION REGIONS

There are over five million farms in the country. Most of these farms are small (81 percent below five ha), accounting for 39 percent of the farmed area. Seven percent of the farms have over 20 ha but occupy 40 percent of the land (Table 1).

The farming systems depend on the land types, micro climate, availability of irrigation water, market access, population density, transport infrastructure and cultural aspects. Most farms are mixed. Farmers in the crop production regions, whether small or large, produce not only the major crops but also minor crops, vegetables

and fruits. They have shade trees for fuel or timber; keep cattle or buffaloes for milk, and own goat/sheep and chickens. Livestock may provide a substantial proportion of the income. There are fifteen crop production regions (CPRs), grouped according to their major cropping patterns. There are five in Punjab, three in Sindh, three in NWFP and four in Balochistan (Figure 3). Details are given in Table 2.

TABLE 1
Farm sizes and areas in Pakistan

Size of farm	Percent of farms	Farmed area (%)
< 5 ha	81	39
5 – 10 ha	12	22
> 20	7	40

Source: Agriculture Statistics of Pakistan, 2002.

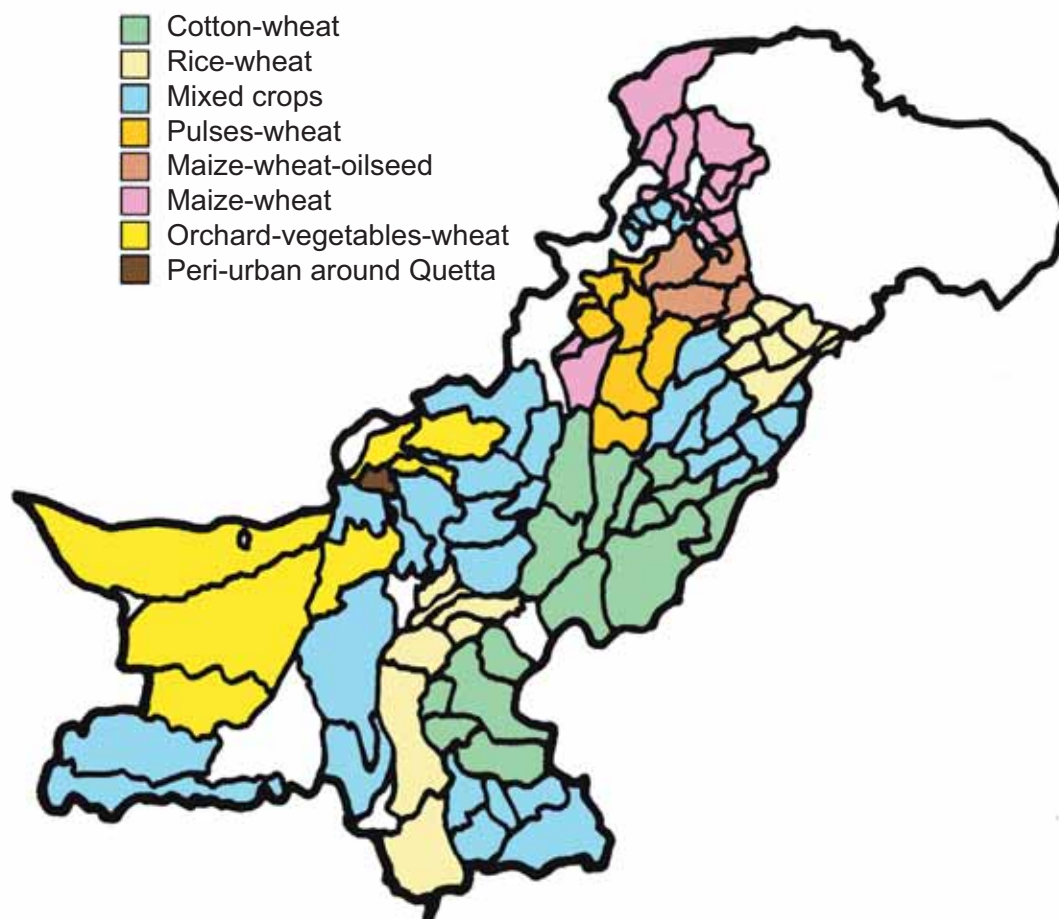
TABLE 2

Crop production regions in Pakistan

No.	Region	Cropping pattern	Agricultural area (million ha)	Source of Irrigation	Rainfall mm (1966–2002)	
					Average	Range
1.	Punjab I	Cotton-wheat	5.5	Canal, tubewell	156	55–247
2.	Punjab II	Rice-wheat	2.8	Canal, tubewell	800	600–1 100
3.	Punjab III	Mixed crops	4.1	Canal, tubewell	446	240–688
4.	Punjab IV	Pulses-wheat	1.9	Canal, rainfed	300	200–550
5.	Punjab V	Maize/wheat-oilseeds	1.2	Rainfed	900	700–1 200
6.	Sindh I	Cotton-wheat	1.6	Canal	50	43–70
7.	Sindh II	Rice-wheat	1.1	Canal	58	40–78
8.	Sindh III	Mixed crops	1.3	Canal, dry	123	62–200
9.	NWFP I	Maize-wheat	0.9	Rainfed	1050	240–1700
10.	NWFP II	Mixed crops	0.53	Canal	520	400–670
11.	NWFP III	Pulses-wheat	0.36	Canal, dry	500	300–600
12.	Balochistan I	Mixed crops	0.40	Tubewell, Karez	180	65–3405
13.	Balochistan II	Orchards/vegetables-wheat	0.30	Tubewell, Karez	115	27–290
14.	Balochistan III	Rice-wheat	0.35	Canal	--	–
15.	Balochistan IV	Peri-urban	0.02	Tubewell, Karez	167	167

Note: Karez = underground water tunnel.

FIGURE 3
Crop production regions (scale = 1:7 000 000)



Source: NFDC.

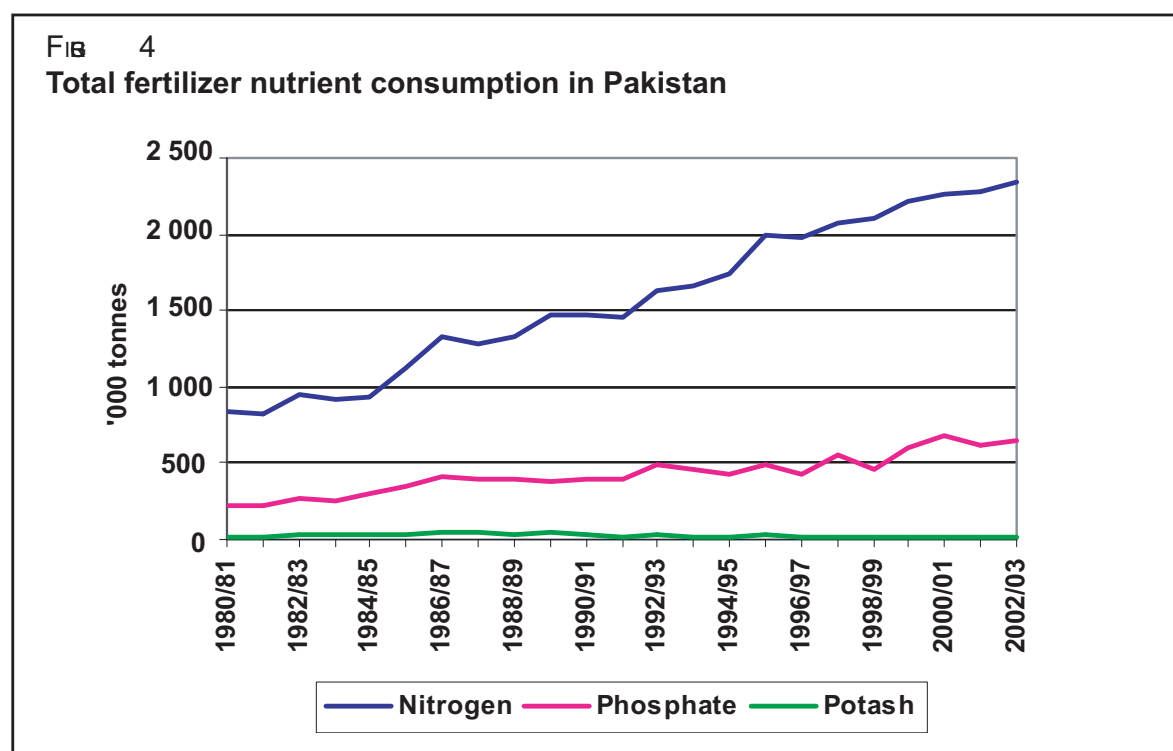
Chapter 3

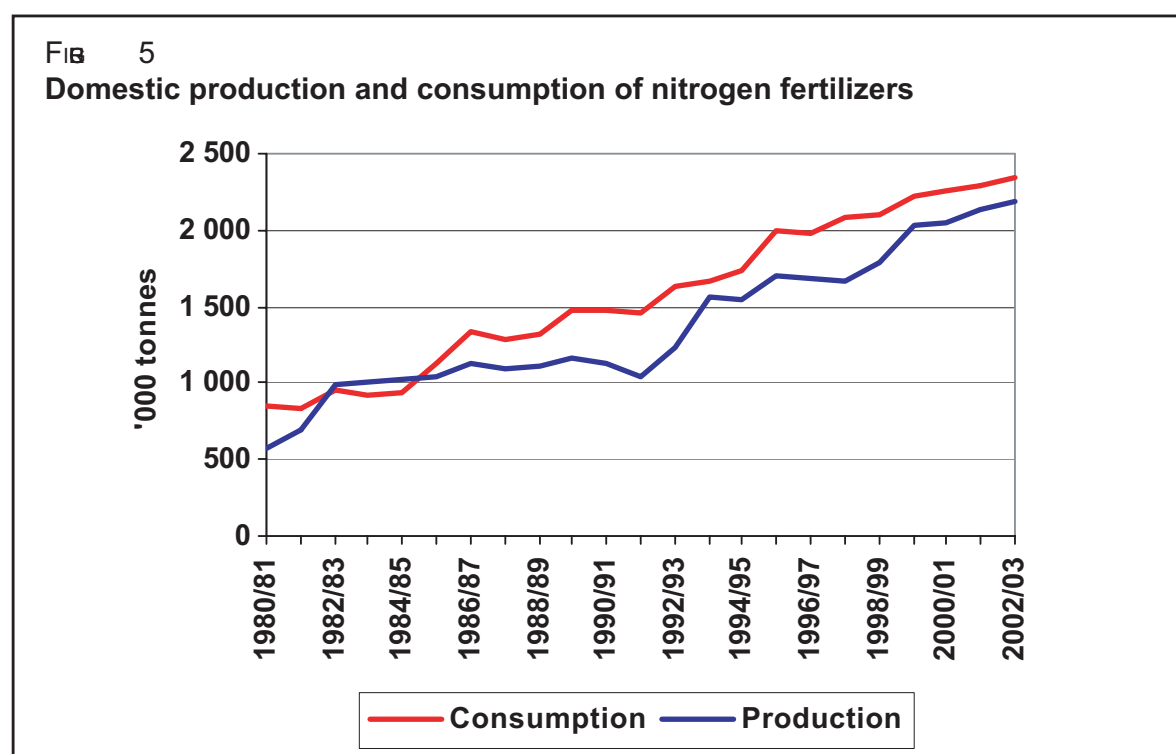
The fertilizer sector

Fertilizer consumption has increased threefold during the past 30 years. It reached one million nutrient tonnes in 1980/81, two million tonnes in 1992/93 and three million tonnes in 2002/03 (Figure 4). Nitrogen accounts for 78 percent of the total nutrients, phosphate for 21 percent and potash for less than one percent. The average N, P₂O₅, K₂O nutrient ratio between 1999/2000 and 2001/02 was 1:0.28:0.01.

FERTILIZER PRODUCTION AND IMPORTS

Fertilizer requirements in the country are met from both domestic production and imports. Currently, there are 14 production units, with a combined design capacity of 2 233 thousand tonnes of N and 239 thousand tonnes of P₂O₅. The annual production capacities are as follows: urea 4.3 million tonnes, DAP 450 thousand tonnes, CAN 450 thousand tonnes, NP compounds 305 thousand tonnes, SSP 180 thousand tonnes and for NPK compounds 100 thousand tonnes





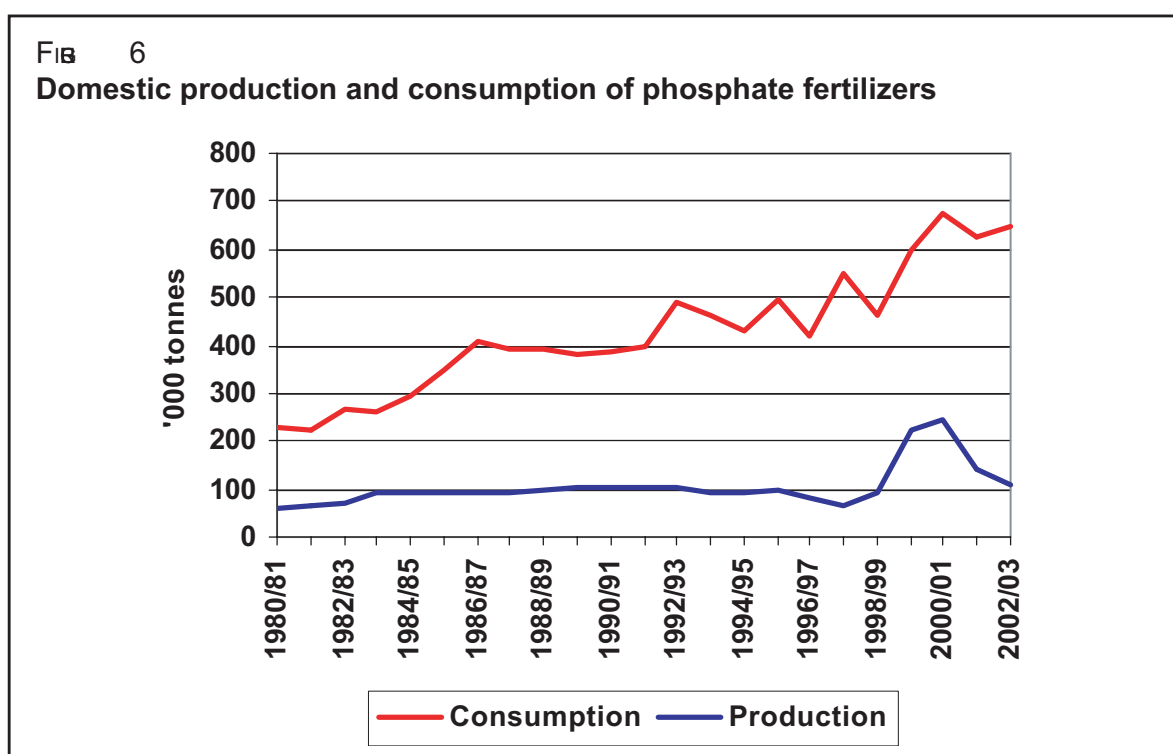
(total 5.78 million tonnes). The overall capacity utilization varies, but it is always more than 100 percent in the case of urea.

Prior to 1980/81, the country imported well over 50 percent of its annual fertilizer requirements. However, the situation changed following the expansion of the local industry during the 1980s.

Figures 5 and 6 illustrate the development of domestic production and consumption of nitrogen and phosphate fertilizers during the period from 1981 to 2003. Nitrogen production has the advantage of domestic natural gas resources. Imports as a percentage of deliveries for the last five years (1998/99 to 2002/03) averaged about 11 percent for nitrogen and 72.5 for phosphate. Urea has not been imported for the last three years; 46 thousand tonnes of urea was exported in 2002/03. The total imports of the different products during the past three years were: DAP 940 thousand tonnes, SOP 18 thousand tonnes, MOP 11 thousand tonnes, AS 23 thousand tonnes, NP 34 thousand tonnes and small quantities of NPKs. Total imports thus amounted to about 1.1 million tonnes.

FERTILIZER DELIVERIES AND CONSUMPTION

The consumption of fertilizers in Pakistan is determined by geography, weather (water availability), prices and the timely availability of the various products.



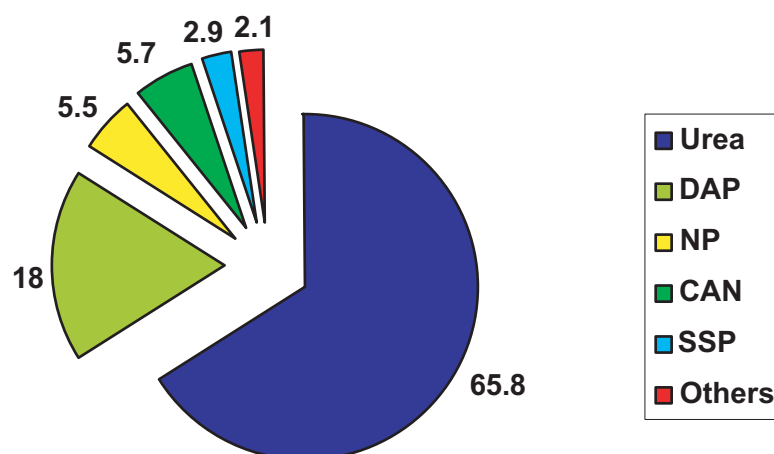
Punjab has the largest agricultural area and therefore consumes the greatest share of fertilizers, followed by Sindh, NWFP and Balochistan, in that order.

Seasonal deliveries are split between Kharif and Rabi seasons. In Kharif 2002 deliveries amounted to 1 384 thousand tonnes of nutrients and in Rabi 2002/03 to 1 634 thousand tonnes of nutrients. On a three year basis the share of urea was 65.8 percent, DAP 18.0 percent, NP compounds 5.5 percent, CAN 5.7 percent and 5 percent of various other fertilizers (Figure 7). November, December and January are the main Rabi months and account for 35.5 percent of the total nutrient deliveries in the year. For Kharif, the main peak months are June, July and August accounting for 32.6 percent of the annual deliveries. On

TABLE 3
Provincial crop areas and fertilizer deliveries 2002/03

Province	Cropped area (million ha)	Percent of total	Fertilizer deliveries ('000 tonnes)	Percent of total
Punjab	16.10	728	2 063	683
Sindh	3.16	144	674	224
NWFP	2.01	90	204	67
Balochistan	0.85	38	77	26
Total	22.12	100	3 019	100

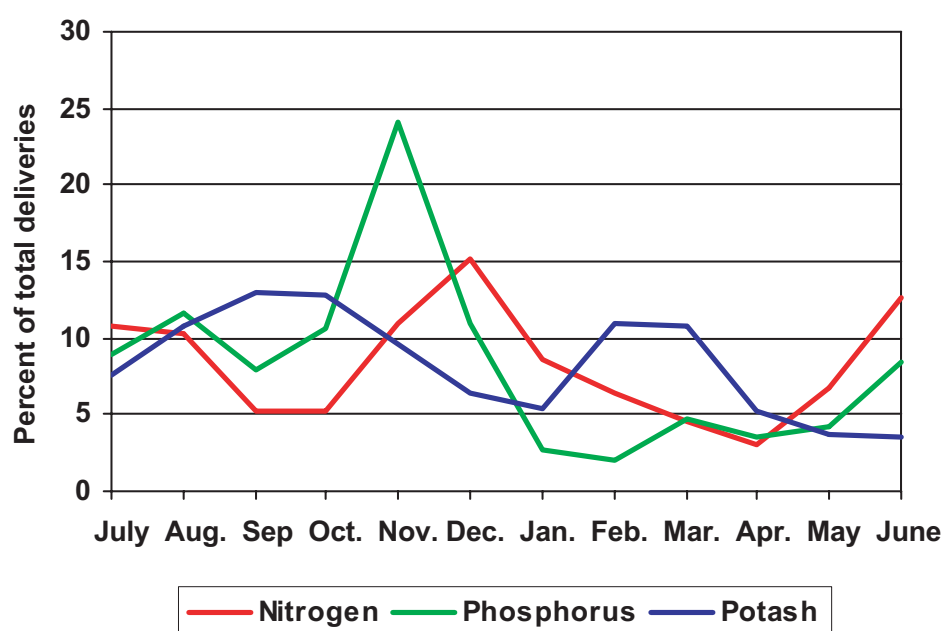
FIG 7

Fertilizer consumption by product (percentages)

a three year basis, the percent of deliveries in each month of N, P_2O_5 and K_2O is presented in Figure 8.

The forecasting of fertilizer demand is effected using econometric as well as agronomic techniques. It is projected that over the next 10 years fertilizer nutrient consumption will grow at the rate of 2 to 3 percent per annum.

FIG 8

Monthly pattern of fertilizer nutrient deliveries (three year average 2000/03); total 2.97 million tonnes

FERTILIZER POLICY

The government has privatized and deregulated fertilizer imports and prices. In 1986, all subsidies on nitrogenous fertilizers were abolished followed by phosphates in 1993 and potash in 1997. Provincial quotas were abolished, provincial supply organizations in the public sector abandoned and import controls were lifted. All imports are effected by the private sector. In 2001, the government imposed a 15 percent general sales tax on all fertilizer products. Farmers have to pay international prices for imported products, apart from urea.

The share of the private sector in fertilizer marketing is 89 percent, compared to 11 percent for the public sector. The private sector handles about 90 percent of the urea and 100 percent of the DAP, the two major fertilizer products consumed in the country. A dealer network of about 8 000 retailers exists in the country. Fertilizer companies select and train the dealers. There is no government intervention. However, under 'Fertilizer Acts' promulgated by provinces, fertilizer quality is monitored by the provincial governments.

Chapter 4

Fertilizer recommendations

Fertilizer recommendations for different crops are prepared by provincial research institutes and released for information to the agriculture extension services for dissemination to the farming community. Federal institutes (PARC, NFDC) and the fertilizer industry also compile information of this kind and publish it for the farmers. NFDC, under an FAO project, developed a methodology for interpreting crop responses to fertilizers, economic analysis and calibration with soil analysis. The general recommendations issued by government departments are given in Table 4.

TABLE 4
Fertilizer recommendations for crops

Crop	CPR	Nitrogen Phosphate Potash		
		(kg/ha)		
Wheat irrigated	Punjab I, II, III	75–160	60–110	60
	Sindh I, II, III	130–170	60–90	50
	NWFP II	120–150	60–90	50
	Balochistan III	90–120	60	50
Wheat rainfed	Punjab V	100	75	60
	Punjab IV	60–75	60	-
	NWFP I	60–100	30–60	-
	Balochistan II	60	30	-
Paddy	Punjab II	75–160	67	60
	Sindh II	134–180	67–100	50
	NWFP I	120–150	90	60
	Balochistan III	120	60	50
Cotton	Punjab I	120–170	60	60
	Sindh I	90–140	60	50
Sugar cane	Punjab III	170–270	60–110	60–120
	Sindh III	200–300	100–125	100–170
	NWFP II	120–175	100	100
Maize	NWFP I	90–120	60–90	30–60
	Punjab V	60–90	60	--
Pulses	Punjab IV	20–30	60–90	--
	Sindh III	90–120	60	--
	NWFP III	25–50	80	--

The general recommendations are mostly given as a range, low for fertile soils and higher rates for fields with poor soil fertility. The previous crop, soil texture, source of irrigation and rainfall are also taken into consideration. There is a network of over 60 soil testing laboratories in both the public and private sectors to provide recommendations based on soil analysis for specific soils and crops.

The amount of fertilizer that should be applied depends on the crop to be sown, the previous crop, the organic manures that will be applied, the crop variety, the input/output price ratio, yield potential and management level. The use of site specific recommendations by farmers is negligible. Farmers apply fertilizers according to their financial resources, the availability of water, the types of fertilizers available and the expected financial returns.

GENERAL GUIDELINES

Wheat

All phosphorus and potassium and half of the nitrogen is broadcast and incorporated in the soil before sowing.

Phosphorus can be applied at the first irrigation if this was not done at sowing.

The remaining half of the nitrogen is top dressed with the first or second irrigation.

On light textured soils, nitrogen should be applied in three splits.

In rainfed areas all fertilizer could be applied at sowing.

In the case of late planting, it is better to apply all the fertilizers at sowing and slightly increase the dose.

Potassium, zinc (Zn) and other micronutrients should be applied where needed based on the results of soil analysis.

Rice (Paddy)

Application of zinc sulphate (35 percent Zn) at the rate of 12.5 kg/ha after 7–10 days of transplanting.

All P_2O_5 and K_2O and half of the N are to be applied before transplanting. Nitrogen fertilizer should not be applied in standing water to avoid volatilization losses; it should be incorporated in the mud. Basal urea incorporated into the mud and later top dressed in the wet soil followed by flooding can increase the yield by 25-30 percent compared with the same amount of urea applied by the common method i.e. fertilizer application in standing water.

The remaining half of the N is applied at the panicle initiation stage i.e. 35 to 40 days after transplanting, in dry soil followed by flood irrigation.

Nitrogen fertilizers containing nitrogen in ammoniacal form (urea, ammonium sulphate) are more beneficial for rice.

Potassium fertilizer should be applied based on soil analysis.

Cotton

All P_2O_5 , K_2O and 1/3 of the N should be applied at sowing by band placement.

Then 1/3 of the N should be applied with the first irrigation and the remaining 1/3 of the N at the pre-flowering stage.

If phosphate was not applied at sowing, it should be top dressed along the lines before the first irrigation.

In case the wheat crop is fully fertilized with phosphorus and the soil test value is above 10.0 mg/kg, then P_2O_5 application to cotton can be reduced.

Potassium and micronutrients (boron) should be applied on the basis of soil analysis. Boron is important for cotton, but one has to be careful because the difference between deficiency and toxicity is very narrow.

Sugar cane

All phosphorus, potassium and 1/3 of the N should be applied at planting time in the furrows below the seed sets. Fertilizer contact with the seed sets has to be avoided.

The remaining 2/3 of the N should be applied in two splits, i.e. 1/3 in April and 1/3 in May.

Twenty to 25 cart loads of farm yard manure (FYM) per hectare are also recommended to be applied at the time of soil preparation at least one month before planting.

For the ratoon crop about 30 percent more fertilizer are required than for the newly planted sugar cane. N should be applied in two splits, i.e. half along with P_2O_5 and K_2O at the time of sprouting and the remaining half two months after the first dose.

Fruit crops

Most of the recommendations for citrus fruits, deciduous fruits and mango are on a per plant basis. After planting the fruit tree, the application in the first year on average is 150 g N, 50 g P_2O_5 and 50 g K_2O , mixed with 10–15 kg of well decomposed FYM for every plant. The dose in g is incrementally increased every year by about 75 g for N, 50 g for P_2O_5 and 25 g for K_2O . Thus a mature plant of about 10 years will receive 1.5 kg N, 0.5 kg P_2O_5 and 0.5 kg K_2O , mixed well with about 50 kg well decomposed FYM. Micronutrients are also to be added.

Chapter 5

Organic and biological sources of plant nutrients

Censuses and surveys show that farmers apply FYM when it is available and when there is adequate labour available to collect, store and spread it. The growing of a leguminous crop such as pulses and groundnuts is limited by the need to grow more profitable crops. Green manuring, especially with dhaincha (*Sesbania aculeata*) and guar bean (*Cyamopsis tetragonoloba*) is fairly widespread in areas where water is available.

The 1990 Agricultural Census data on farmers' current use of manures and fertilizers show that 32.4 percent of farmers applied some form of organic plant nutrients on maize and 29.3 percent on sugar cane. A substantial number of farmers applied organic nutrients on tobacco, potatoes and fruit crops. Even on wheat, nearly 20 percent of the farmers applied organic sources of plant nutrients.

A survey conducted by NFDC (2000) shows that major nutrients like nitrogen and phosphorus are the most commonly used fertilizers followed by FYM and micronutrients. About 50 percent of the farmers reported that they use FYM on one crop or another.

ANIMAL WASTES AND FARM YARD MANURE

Pakistan has a huge population of livestock. It has been estimated that about 50 percent of animal wastes are not collected. About 50 percent of the quantity recovered is used as fuel. Thus scarcely a quarter of the animal wastes are available for use as organic sources of plant nutrients. The animal wastes together with an equal quantity of stable bedding material, left-over fodder and household wastes provide the total quantity of FYM available. Based on different assumptions, it is estimated that about 1.5 million tonnes of nutrients are available from FYM. Of this quantity, nitrogen accounts for 726 thousand tonnes, P_2O_5 for 191 thousand tonnes and K_2O for about 617 thousand tonnes.

POULTRY MANURE

Poultry manure is rich in nutrients. The estimates show that if poultry manure is properly managed it can contribute about 101 thousand tonnes of nitrogen, 58 thousand tonnes of P_2O_5 and 26 thousand tonnes of K_2O .

CROP RESIDUES

A huge quantity of crop residues such as wheat straw, cotton stems, sugar cane trash/tops and rice husks, is available. But due to economic necessities such as the need for animal fodder and fuel, the crop residues cannot be recycled in the soil. Kallar grass (*Leptochloa fusca*) is recognized as a salt tolerant grass capable of producing a good amount of biomass on degraded soils in summer.

FILTER CAKE AND STILL EFFLUENT

The Pakistan sugar industry produces over 1.2 million tonnes of filter cake every year, which is a rich source of organic matter, micro and macronutrients. Some sugar mills have molasses-based distillery plants, which produce still effluent that contains nutrients, especially potassium. Most of the filter cake is sold to the brick baking industry and the still effluent is drained out, with a consequent loss of plant nutrients and environmental pollution.

OTHER SOLID AND LIQUID BASED MATERIALS

Other solid and liquid based materials available include sewage and sludge, waste water, fish pond effluent, city refuse and some wastes from food processing industries. These materials cannot be used directly as a source of plant nutrients. However, after proper processing and removal of heavy metals and other undesirable material, these sources have the potential of being a good source of plant nutrients.

BIOLOGICAL SOURCES

There is a good potential for quality-controlled biofertilizers, such as those based on nitrogen-fixing bacteria, to contribute to soil fertility, supplementing inorganic fertilizers.

Chapter 6

Fertilizer use by crop

The total cropped area in Pakistan is about 22.2 million ha. The share of food grain crops is 54 percent, followed by cotton and sugar cane 20 percent; pulses 6 percent; oilseed crops 3 percent; fruit/vegetables 4 percent and other crops about 13 percent. Figure 9 shows that wheat is the main food crop. It occupies about 36.3 percent of the total cropped area, followed by cotton with 14 percent, paddy with 9.5 percent, sugar cane with 4.5 percent, maize with 4.5 percent and other crops with 20.8 percent.

According to the fertilizer use survey five major crops: wheat, cotton, sugar cane, rice and maize account for about 87 percent of fertilizer consumption. Wheat accounts for about 45 percent followed by cotton with a share of 23 percent. Sugar cane is the third crop; nutrient use per ha is highest on this crop. The share of fruit and vegetables is 5.6 percent (Figure10).

The yields of the major crops (Table 5) are below their agronomic and genetic potential. There is a consensus among researchers, extensionists and policy planners that, given the necessary resources and inputs, yields could be increased by 30 to 40 percent.

FIGURE 9

Percentage of cropped area under different crops

Total area 22.2 million ha

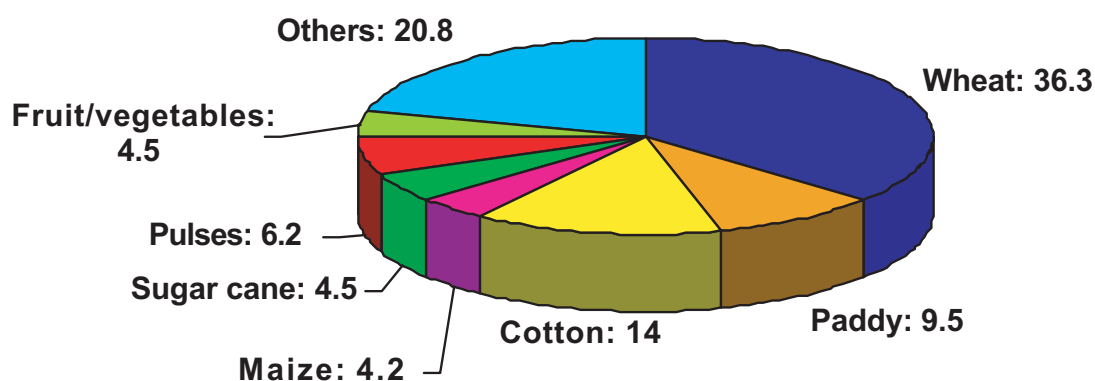
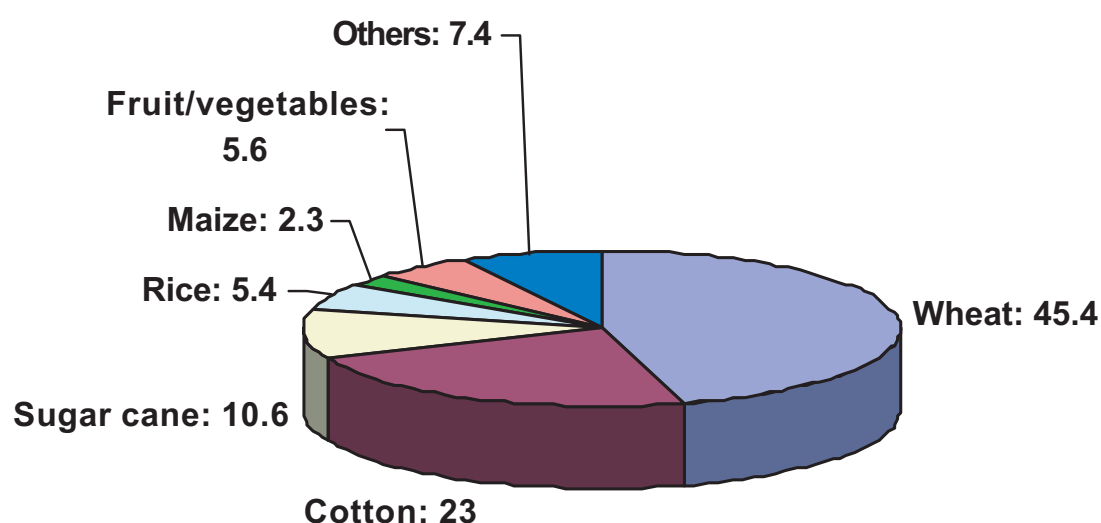


FIGURE 10

Fertilizer use by crop (percentage)**Total consumption 3 million tonnes****FERTILIZER USE ADOPTION AT FARM LEVEL**

Less than two percent of the farmers apply potash whereas 92 percent apply nitrogen and 83 percent apply phosphate. Half of the farmers apply FYM. Almost five percent apply micronutrients. The use of green manures, crop residues and bio-fertilizers is negligible (Figure 11).

Almost all farmers having access to canal or tubewell irrigation water use fertilizers. In rainfed areas scarcely 50 percent of the farmers use fertilizers. About 92 percent of small-scale farmers (< 5 ha) use fertilizers. Application rates

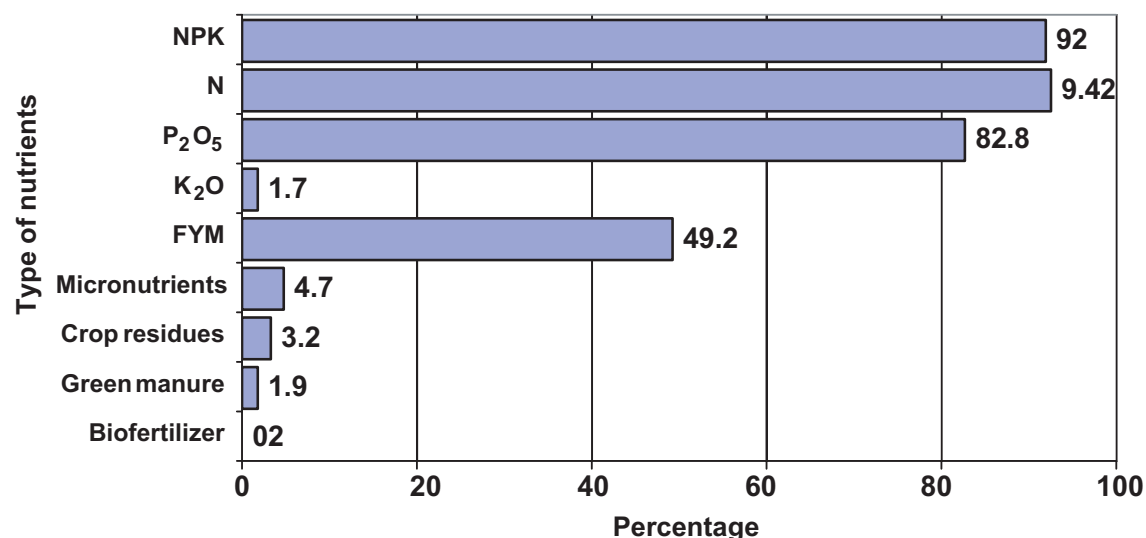
TABLE 5

Yields of major crops in Pakistan

No.	Province	Wheat		Cotton	Sugar cane	Rice	
		Irrigated	Rainfed			IRRI	Basmati
(kg/ha)							
1.	Punjab	2 562	717	542	48 400	1 928	1 479
2.	Sindh	2 417	1 196	759	47 400	2 265	--
3.	NWFP	1 809	734	436	47 200	2 165	1 985
4.	Balochistan	2 031	291	502	50 000	3 037	2 390
5.	Pakistan	2 487	721	579	48 100	2 539	1 501

Source: Agriculture Statistics Pakistan 2001–02.

FIGURE 11

Adoption of different sources of plant nutrients by farmers

decrease with an increase in farm size (rainfed and irrigated). In rainfed areas the difference between application rates for small and large farms is very wide.

FERTILIZER APPLICATION RATES BY CROP

The crop-wise per ha use of fertilizer generally varies with the farm size, sources of irrigation, educational level, land tenure and cropping system. NFDC occasionally conducts farm level surveys to assess fertilizer use by crop and its impact on productivity. The other source of information is fertilizer deliveries in a specific administrative or ecological zone from which use per hectare can be calculated. To arrive at the figures of use by crop, all the information available through surveys and fertilizer deliveries has been used. The use on crops in each crop production zone is given in Table 6.

A comparison between fertilizer recommendations (Table 4) and actual use (Table 6) shows that the nitrogen application rate is close to 80 percent of the recommendations, compared with about 40 percent or less, depending on the crop, in the case of phosphate. Rates of nitrogen application approach recommended rates on irrigated wheat and cotton but are well below the recommended rates in the case of rainfed wheat. Scarcely two percent of the farmers apply potash, the small quantities used are applied mainly on sugar cane

and vegetables and fruits. The $N:P_2O_5:K_2O$ ratio during the past three years has averaged 1:0.28:0.01.

TABLE 6

Fertilizer use by crop in the CPRs

CPR	Wheat				Cotton		Sugar cane		Rice			
	Irrigated		Rainfed						IRRI		Basmati	
	N	P ₂ O ₅	N	P ₂ O ₅	N	P ₂ O ₅	N	P ₂ O ₅	N	P ₂ O ₅	N	P ₂ O ₅
	(kg/ha)											
Punjab I	110	40			150	20	150	30			75	18
Punjab II	75	25					120	20	75	10	90	20
Punjab III	90	30			150	15	170	30	80	15		
Punjab IV	80	25	30	8								
Punjab V	70	20	20	5								
Sindh I	120	35			160	30	180	30	100	20		
Sindh II	110	30					170	30	120	30		
Sindh III	90	30			140	25	160	25	110	20		
NWFP I	75	25	40	10					100	20		
NWFP II	110	40					140	25	100	20		
NWFP III	70	10	25	2								
Balochistan I	65	8	25									
Balochistan II	60	7	20									
Balochistan III	75	20							70	10		

Chapter 7

Fertilizer prices and profitability of fertilizer use

During the introductory stage in the 1950s, the focus was on introducing and encouraging the use of fertilizers through simple fertilizer trials and demonstrations on farmers' fields and by subsidizing fertilizer prices. Retail prices were fixed by the Government and were kept uniform throughout the country. With the increase in the level of fertilizer use, the emphasis changed to a more balanced use of fertilizer nutrients. However, as the subsidy burden increased, the Government started to phase out the subsidy under the Structural Adjustment Programme and economic reforms. In 1986, all subsidies on nitrogenous fertilizers were removed followed by phosphate fertilizers in 1995 and potassium fertilizers in 1997. Import controls were lifted, the government stopped importing and the private sector took over. The Government imposed a 15 percent general sales tax on all fertilizers in 2001, thus increasing the prices.

FERTILIZER PRICES

Pakistan is self sufficient in urea; hence no urea is imported. In April 2004 the farm gate price of domestically produced urea was about Rs.430 per bag of 50 kg whereas the farm gate price of imported urea would have been around Rs.600 per bag of 50 kg. The price of DAP, the second largest fertilizer product used after urea, is very volatile and depends on international trends. Price disparities lead to high use of urea, and thus to imbalanced fertilizer use at farm level.

Farmers' decisions on how much fertilizer to use for a specific crop, are linked with commodity prices. Higher fertilizer prices combined with lower crop prices can cause farmers to use low rates of fertilizers. The economics of fertilizer use have always been the over-riding consideration in the demand for fertilizers by farmers, especially the relationship between the commodity price and the input price. Support prices of the major crops have not been revised during the past three years. However, farmers have become so dependent on fertilizers for their crop production that they have been left with no option but to increase fertilizer use. Despite this, fertilizer use is still profitable. Recently

TABLE 7

Relative fertilizer and crop prices

Relative fertilizer and crop prices					
Year	Average nutrient prices (Rs./kg)	Wheat	Paddy (Basmati)	Sugar cane	Seed cotton
2000/01	17.60	300	385	40	725
2001/02	19.04	300	385	40	780
2002/03	19.35	300	385	40	800
kg commodity required to purchase one kg of fertilizer nutrient					
2000/01		2.35	1.80	17.60	0.97
2001/02		2.54	1.98	19.04	0.98
2002/03		2.58	2.0	19.35	0.97

TABLE 8

Retail prices of fertilizer products (Rs./50 kg bag)

Year	Urea	DAP	CAN	NP 23-23	SSP (granular)	SSP (powder)	1 US\$ in Rs.
1998/99	346	665	231	457	234	193	46.8
1999/00	324	632	228	466	287	243	51.8
2000/01	363	669	233	468	253	214	58.4
2001/02	392	710	267	518	279	234	61.4
2002/03	412	765	282	539	289	244	58.7

Source: NFDC.

the market prices of cotton and sugar cane have been higher than the support price. The relative prices are given in Table 7.

The retail prices of major fertilizer products during the last five years are given in Table 8. The main increase was in the DAP price due to a high price on the international market. The international price was around US\$220/tonne f.o.b. in March 2004, and in consequence the domestic price reached Rs.1 000/bag of 50 kg.

PROFITABILITY OF FERTILIZER USE

As more fertilizer is applied, the additional quantity of crop yield produced with each successive unit of fertilizer becomes smaller, until a maximum yield is obtained. This is a result of the law of diminishing returns. Thus there is a point on the curve of the response of the crop to fertilizer application at which the additional use of fertilizer either produces no extra yield or may even have a

negative effect. A farmer is interested in profitable returns rather than maximum returns. This implies that the maximum yield per ha does not always mean the highest profit because of greater cost of fertilizers.

An important criterion for the measurement of profitability is the value cost ratio (VCR). This is the ratio between the value of the additional crop yield and the cost of the additional fertilizer.

$$\text{VCR} = \frac{\text{Value of increased yield obtained}}{\text{Cost of additional fertilizer used}}$$

The VCR is the rate of return on the money spent on fertilizers. If the VCR is greater than one, fertilizer use is profitable. A VCR of 2 represents a 100 percent return on the money invested in fertilizer. In general a VCR of 2 ensures good return and takes account of the risk of bad weather and other external factors.

NFDC, in joint collaboration with FAO and the World Phosphate Institute (IMPHOS), has conducted a number of field trials/demonstrations on farmers' fields in the 15 CPRs over the past seven years. A simple economic analysis has been done to show economic impact of balanced use of fertilizers on various crops. The VCR of the major crops in the CPRs is given in Table 9.

TABLE 9
VCRs of major crops in different CPRs

No.	CPR	VCR				
		Wheat		Paddy		Cotton
		Rainfed	Irrigated	Basmati	IRRI	
1.	Punjab I		3.3			4.1
2.	Punjab II		3.9	4.0		
3.	Punjab III		4.0	7.0		
4.	Punjab IV	4.0				
5.	Punjab V					
6.	Sindh I		3.8		2.5	11.0
7.	Sindh II		3.7		1.5	10.0
8.	Sindh III		3.6		3.3	
9.	NWFP I		3.5			
10.	NWFP II	3.0	3.7			
11.	NWFP III		2.7			
12.	Balochistan I		5.0			
13.	Balochistan II					
14.	Balochistan III				3.7	
15.	Balochistan IV					

Prices used: N = Rs.18/kg, P₂O₅ = Rs.32/kg, wheat = Rs.8.75/kg, cotton Rs.25/kg, Basmati rice Rs.11.50/kg, IRRI rice Rs.6/kg and maize Rs.11/kg.

The results show that fertilizer use remains profitable provided it is balanced (N, P₂O₅). However, the present application rates particularly of phosphate are low. Thus the present VCR obtained by farmers ranges between 2 to 3.

Chapter 8

Fertilizer research and development

Research on fertilizer use in Pakistan was initiated in 1909, with the establishment of the Punjab Agriculture College and Research Institute at Faisalabad (then Lyallpur), followed by the establishment of various research stations in the country. The response of various crops to nitrogen containing fertilizers was published in 1934. Phosphorus deficiency was reported for the first time in 1952. A Soil Fertility Research and Fertilizer Popularizing Organization was set up under FAO auspices in 1958, with the mandate to conduct applied fertilizer research and promote fertilizer use. Under this programme intensive experimentation was carried out on farmer's fields to create awareness about the role of nitrogen, phosphorous and potassium in crop production. This led to the use of fertilizers at farm level and also to investment in the national fertilizer industry. The deficiency of zinc in rice was identified in 1969 and the deficiency of boron in cotton was reported in 1970. Horticultural crops, particularly deciduous fruits, showed a severe deficiency of iron. The importance of zinc for rice is known to rice growers. The Provincial Soil Fertility Institutes, the Pakistan Agricultural Research Council and other national institutes had generated a great deal of information over time.

With the identification of multiple nutrient deficiencies, the focus shifted to balanced fertilizer use, integration of mineral fertilizer with organic sources, quantification of residual nutrients, fertilizer recommendations in cropping systems and soil test crop response calibration for economic and optimum fertilizer use based on soil analysis.

As the fertilizer sector moved towards maturity, issues related to policy and economics emerged. Therefore, in 1978 the Government of Pakistan with the assistance of FAO established the National Fertilizer Development Centre (NFDC) with the mandate to study all issues of the fertilizer sector from source of supply to farmer's fields, with a view to developing policy advice and to assist, where appropriate, in its implementation. Its scope covered policy planning, marketing and statistics, monitoring of supply and demand and fertilizer research, training, and extension activities. NFDC played a central role in the field of policy issues. It analyzed the role of the public and private fertilizer sector, reserve stocks, phasing out of subsidies and servicing of remote areas to

facilitate privatization and deregulation. Short and long term fertilizer demand forecasts, pricing, fertilizer production, import and marketing were other areas under consistent review. Studies were conducted on fertilizer use at farm level, its impact on crop productivity, the quality of products and constraints in balanced and efficient use of fertilizers.

The data on crop response to fertilizers were collected, collated, analyzed and interpreted. Fertilizer research trials on farmer's fields were harmonized and deficiencies were identified in soil testing laboratories. The topics of fertilizer use efficiency, economic optimum recommendations, and integrated plant nutrient management were also addressed. With support from IMPHOS and FAO, data were obtained on the balanced use of fertilizer and its contribution to crop production. NFDC maintains a comprehensive data base and issues monthly status reports on fertilizer production, imports, prices, use at farm level and the issues for policy action. NFDC also coordinates with federal and provincial research institutes, extension and fertilizer industry through providing a common platform to discuss and deliberate on contentious issues.

The objective of fertilizer research and development has shifted to the improvement of efficiency, increased crop productivity and minimization of the impact on the environment. The overall aim is that sustainability in agricultural growth should match the growing population for food security and the promotion of economic growth.

Chapter 9

Constraints

The constraints on agricultural productivity include soil degradation (soil salinity, erosion and soil fertility depletion), depletion of water resources, mismanagement of irrigation systems, the distribution of the land holdings and poor farming practices. The use of farm inputs, particularly of fertilizers, is inadequate and inefficient. Farm energy use is low. The availability of quality seed is limited and the use of fertilizers is imbalanced.

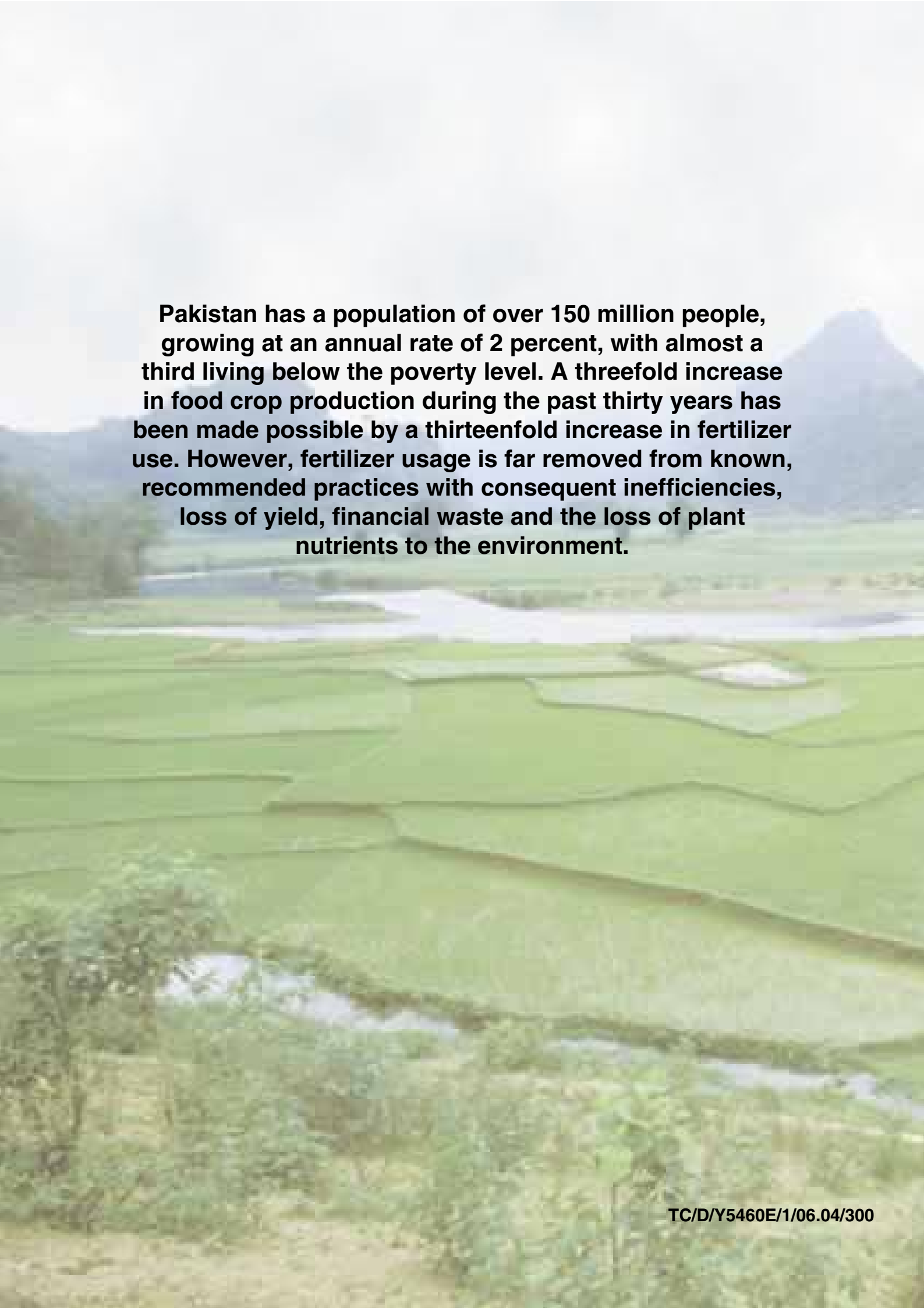
Agricultural research is lagging behind the new challenges. Agricultural extension services are not tuned to modern technology. The flow of information from research to farmers is inadequate. Coordination between policy, research, extension and farmers could be improved. Disbursement of agricultural credit amounts to over Rs.50 billion per annum, but is less than the requirements and is not reaching small farmers. An inadequate marketing infrastructure results in high marketing costs and losses.

Expansion of domestic fertilizer production, particularly of phosphate and NPK compounds is not envisaged in the near future. In consequence, imbalances in the nutrient ratios may continue. Micronutrient deficiencies are affecting yields in many parts of country. The fertilizer recommendations are too general. Soil testing laboratories are not adequately equipped in terms of manpower and equipment.

References

- Ahmad, N.** 1999. Fertilizer scenario in Pakistan: policies and development. In: *Proc. Conference on Agriculture and Fertilizer Use by 2010*. NFDC Publication 2/99. Islamabad.
- Ahmad, N. & Chaudhry, A. G.** 2000. Fertilizer use at farm level. *NFDC publication* 4/2000. Islamabad.
- Ahmad, N.** 2000. Integrated plant nutrition management in Pakistan: status and opportunities. In: *Proc. Symposium on Integrated Plant Nutrition Management*, NFDC. pp:18-37. Islamabad.
- Ahmad, N. & Rashid, M.** 2003. *Fertilizers and their use in Pakistan*. Extension Guide. Islamabad.
- Ahmad, N. & Muhammad, T.** 1998. Fertilizer plant nutrient management and self reliance in agriculture. *The Pakistan Development Review* 37:4, 217–233.
- FAO/UNESCO.** 1974. *Soil Map of the World* 1:5 000 000. Volume I. Legend. UNESCO, Paris.
- FAO.** Web Site: <http://www.fao.org/ag/aquastat>. Rome.
- Government of Pakistan.** 1994. *1990 Census of agriculture*. Agricultural Census Organization. Economic Affairs and Statistics Division. Islamabad.
- Government of Pakistan.** 2001. Water resources development. Ten year perspective development plan. Planning Commission. pp 277–292. Islamabad.
- Government of Pakistan.** 2003. *Agricultural statistics: 2001-02* Ministry of Food, Agriculture and Livestock. Islamabad.
- Government of Pakistan.** 2003. *Pakistan Economic Survey 2002-03*. Ministry of Finance. Islamabad.
- Mian, A. & Javed, Y.** 1993. The soil resources of Pakistan – their potential, present stage and strategies for conservation. *Sector paper for National Strategy*. 53 pp. Islamabad.
- NFC-National Fertilizer Corporation of Pakistan Ltd.** 1981. *A study on differential impact of fertilizer price increase on its use in Pakistan: Fertilizer use pattern in Pakistan Rabi 1980-81*. NFC-NFDC. Islamabad.

- NFDC.** 1989. Crop response to fertilizer. *NFDC Publication 4/89*. Islamabad.
- NFDC.** 1998. Micronutrients in agriculture. *Pakistan Perspective Publication No. 4/98*. Islamabad.
- NFDC.** 2000. Fertilizer use survey. *NFDC Publication*. Islamabad.
- NFDC.** 2001. Balanced fertilization through phosphate promotion. *FAO/IMPHOS/NFDC Report*. Islamabad.
- NFDC.** 2003. *Fertilizer Review 2002-03*, National Fertilizer Development Centre (NFDC), Planning and Development Division. Islamabad.
- NFDC.** National Fertilizer Development Centre. 1996. *Fertilizer use on wheat: farm level survey Rabi 1991-92*. Mimeo. Islamabad.
- PARC.** 1980. *Crop ecological regions in Pakistan*. Memograph. Islamabad.
- Twyford, I. T., Ghaffar Chaudhry, M. & Muhammad Sadiq.** 1995. *Plant nutrition policies and strategies for sustainable agricultural development in Pakistan: executive summary, policy options and action programme*. National Fertilizer Development Centre. Islamabad.
- Twyford, I. T., Bajwa, M. I. & Tahir Saleem, M.** 1993. *Fertilizer use at farm level in Pakistan*. National Fertilizer Development Centre. Islamabad.



Pakistan has a population of over 150 million people, growing at an annual rate of 2 percent, with almost a third living below the poverty level. A threefold increase in food crop production during the past thirty years has been made possible by a thirteenfold increase in fertilizer use. However, fertilizer usage is far removed from known, recommended practices with consequent inefficiencies, loss of yield, financial waste and the loss of plant nutrients to the environment.